

March 18th Game Theory Notes

Jacob Wyngaard

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1 Best Response Polytopes with Negative Expectation

Recall that for a two player game with payoff matrices Π^1 and $\Pi^2 \in \mathbb{R}^{m \times n}$, the best response polytopes are defined as:

$$P := \{p^1 \in \mathbb{R}^m : p^1 \geq 0, (\Pi^2)^T p^1 \leq \mathbf{1}\}, \quad Q := \{p^2 \in \mathbb{R}^n : p^2 \geq 0, (\Pi^1)p^2 \leq \mathbf{1}\}.$$

The construction of these polytopes assume that the expected value for either player at the Nash equilibrium is 1, which reflects a scaling that is passed on to the probability vectors. However, the expected value could be negative. For example, in the game defined by the payoff matrices below, Player 2 has a negative expectation value:

$$\Pi^1 = \begin{pmatrix} 6 & -2 & 3 \\ -4 & 5 & 4 \end{pmatrix}, \quad \Pi^2 = \begin{pmatrix} -6 & 2 & -3 \\ 4 & -5 & -4 \end{pmatrix}.$$

In this case, the Player 1 polytope P generated by the constraints $p^1 \geq 0$ and $(\Pi^2)^T p^1 \leq \mathbf{1}$ does not contain the Nash equilibrium as a vertex. However, the Nash equilibrium does live as a vertex on the polytope $\{p^1 \in \mathbb{R}^m : p^1 \geq 0, (\Pi^2)^T p^1 \leq -\mathbf{1}\}$. This polytope can be thought of as being generated by scaling the probability vector by the absolute value of Player 2's expected value at the Nash equilibrium.

The question remains regarding when we generate our polytope using constraint $(\Pi^2)^T p^1 \leq \mathbf{1}$ and when to use constraint $(\Pi^2)^T p^1 \leq -\mathbf{1}$. So far, the intuition appears to be that, if the polytope generated has a vertex with all positive components, then that polytope is the correct one. The proof of this is an active endeavor.

2 Brown-von Neumann-Nash Dynamical System

2.1 Game Setup

Consider a two-player game with payoff matrices:

$$\Pi^1 = \begin{pmatrix} 7 & 2 \\ 3 & 4 \end{pmatrix}, \quad \Pi^2 = \begin{pmatrix} 1 & 6 \\ 8 & 2 \end{pmatrix}.$$

Player 1 uses strategy

$$p^1 = (x^1, y^1), \quad x^1 + y^1 = 1,$$

and Player 2 uses

$$p^2 = (x^2, y^2), \quad x^2 + y^2 = 1.$$

2.2 Expected Payoffs

The expected payoff for Player 1 is:

$$u^1(p) = 7x^1x^2 + 2x^1y^2 + 3y^1x^2 + 4y^1y^2.$$

The expected payoff for Player 2 is:

$$u^2(p) = x^1x^2 + 6x^1y^2 + 8y^1x^2 + 2y^1y^2.$$

2.3 Excess Payoff Functions

Define the excess payoff (positive deviation from current expected payoff):

Player 1

$$\begin{aligned}\varphi_{11}(p) &= \max(0, 7x^2 + 2y^2 - u^1(p)), \\ \varphi_{12}(p) &= \max(0, 3x^2 + 4y^2 - u^1(p)).\end{aligned}$$

Player 2

$$\begin{aligned}\varphi_{2A}(p) &= \max(0, x^1 + 8y^1 - u^2(p)), \\ \varphi_{2B}(p) &= \max(0, 6x^1 + 2y^1 - u^2(p)).\end{aligned}$$

2.4 BNN Dynamics

The Brown–von Neumann–Nash dynamics are given by:

$$\dot{p}_{ij} = \varphi_{ij}(p) - p_{ij} \sum_k \varphi_{ik}(p).$$

Expanded System

Player 1

$$\begin{aligned}\dot{x}^1 &= \varphi_{11}(p) - x^1(\varphi_{11}(p) + \varphi_{12}(p)), \\ \dot{y}^1 &= \varphi_{12}(p) - y^1(\varphi_{11}(p) + \varphi_{12}(p)).\end{aligned}$$

Player 2

$$\begin{aligned}\dot{x}^2 &= \varphi_{2A}(p) - x^2(\varphi_{2A}(p) + \varphi_{2B}(p)), \\ \dot{y}^2 &= \varphi_{2B}(p) - y^2(\varphi_{2A}(p) + \varphi_{2B}(p)).\end{aligned}$$

2.5 Equivalent Replicator-Like Form

Using $x^1 + y^1 = 1$ and $x^2 + y^2 = 1$, we can rewrite:

Player 1

$$\begin{aligned}\dot{x}^1 &= y^1 \varphi_{11}(p) - x^1 \varphi_{12}(p), \\ \dot{y}^1 &= x^1 \varphi_{12}(p) - y^1 \varphi_{11}(p).\end{aligned}$$

Player 2

$$\begin{aligned}\dot{x}^2 &= y^2 \varphi_{2A}(p) - x^2 \varphi_{2B}(p), \\ \dot{y}^2 &= x^2 \varphi_{2B}(p) - y^2 \varphi_{2A}(p).\end{aligned}$$

2.6 Simulation Results

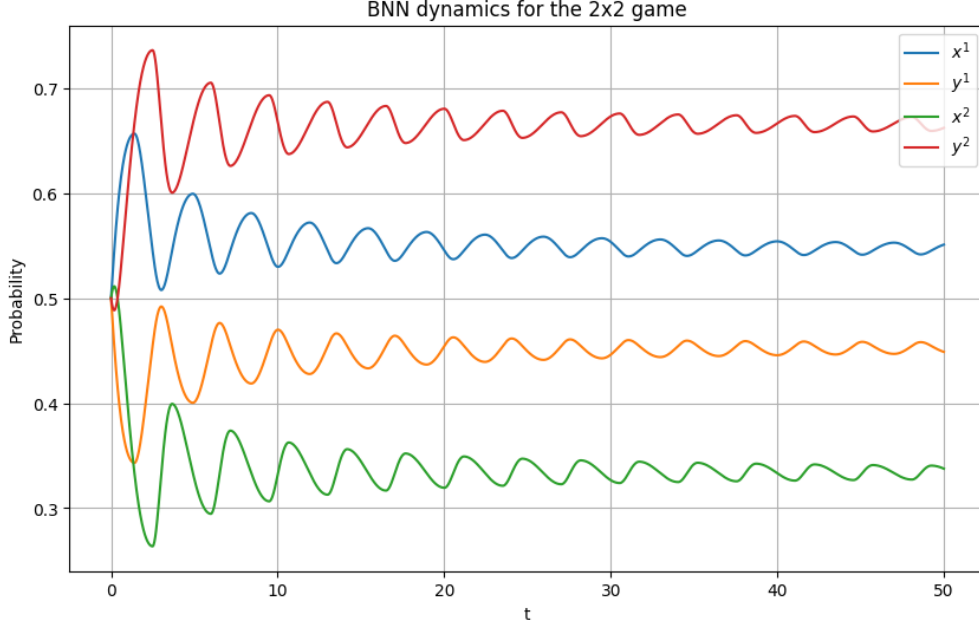


Figure 1: Simulation Results

These are the ODE simulation results obtained using an RK45 method.

Note that the initial conditions were

$$x^1 = y^1 = x^2 = y^2 = \frac{1}{2}.$$

Furthermore, the Nash equilibrium of this system is given by

$$(x^1, y^1) = \left(\frac{6}{11}, \frac{5}{11}\right),$$

$$(x^2, y^2) = \left(\frac{1}{3}, \frac{2}{3}\right).$$

3 Fictitious Play

3.1 Introduction

Fictitious play is a discrete-time analogue of Brown–von Neumann–Nash (BNN) dynamics, originally introduced by Brown (1951) and further studied by Robinson. It is also referred to as *Brown–Robinson learning*.

In this framework, the excess payoff function $\varphi_{ij}(p)$ is replaced by an *ordinal potential* function $F(i, j)$, which preserves preference orderings. Specifically, F satisfies:

$$F(i, j) > F(k, j) \iff \Pi_{ij}^1 > \Pi_{kj}^1, \quad (1)$$

$$F(i, j) > F(i, k) \iff \Pi_{ij}^2 > \Pi_{ik}^2. \quad (2)$$

Thus, higher potential values correspond to higher payoffs, and the function F encodes only the ranking of strategies rather than their exact numerical values.

3.2 Set-Up

Fictitious play can be implemented in two ways:

- Alternating fictitious play (AFP)
- Simultaneous fictitious play (SFP)

While SFP has been more extensively studied, global convergence results in nondegenerate ordinal games are currently established only for AFP.

Let p_k^1 and p_k^2 denote the mixed strategies for Players 1 and 2 at iteration k . These may be initialized arbitrarily (e.g., uniformly or as pure strategies).

At each iteration:

- Player 1 plays a pure strategy best response to p_k^2 :

$$e_{k+1}^1 = \arg \max_{s^1} (s^1)^\top \Pi^1 p_k^2.$$

- Player 2 plays a pure strategy best response to p_{k+1}^1 :

$$e_{k+1}^2 = \arg \max_{s^2} (p_{k+1}^1)^\top \Pi^1 s^2.$$

The mixed strategies are then updated via averaging:

$$p_{n+1}^i = \frac{1}{n+1} \sum_{k=1}^{n+1} e_k^i \tag{3}$$

$$= \frac{n}{n+1} p_n^i + \frac{1}{n+1} e_{n+1}^i, \quad i = 1, 2. \tag{4}$$

3.3 Degenerate Games

A game is said to be *degenerate* if at least one player has multiple best responses to a given opponent strategy. Formally, degeneracy occurs if:

- There exist $i, j, k \neq i$ such that $\Pi_{ij}^1 = \Pi_{kj}^1$, or
- There exist $i, j, k \neq j$ such that $\Pi_{ij}^2 = \Pi_{ik}^2$.

In other words, degeneracy arises when payoff ties occur between distinct pure strategies.

3.4 Convergence of Fictitious Play

Theorem 1 (Berger, 2005). *Let (Π^1, Π^2) correspond to a nondegenerate ordinal potential game. Then every alternating fictitious play (AFP) process converges to a pure Nash equilibrium.*

3.5 Proof Sketch (by Contradiction)

Suppose the AFP process does not converge. Then there exists a pair of non-equilibrium pure strategies (i, j) that are played infinitely often.

Since (i, j) is not a Nash equilibrium, at least one player has a profitable deviation:

- Either Player 1 has a strategy $k \neq i$ that is a better response to j , or
- Player 2 has a strategy $\ell \neq j$ that is a better response to i .

However, fictitious play ensures that players repeatedly best respond, and expected payoffs improve over time. I am currently working on explicitly proving that the expected payoffs improve over time. Returning infinitely often to (i, j) would imply that a unilateral deviation does not improve payoff, contradicting the assumption that (i, j) is not a Nash equilibrium. Thus, the process must converge to a pure Nash equilibrium.

4 BNN Modeling Python Code

```
1 import numpy as np
2 from scipy.integrate import solve_ivp
3 import matplotlib.pyplot as plt
4
5 # -----
6 # Payoff matrices from the slides
7 # Player 1 payoff matrix:
8 #      A B
9 # 1 [ 7, 2 ]
10 # 2 [ 3, 4 ]
11 #
12 # Player 2 payoff matrix:
13 #      A B
14 # 1 [ 1, 6 ]
15 # 2 [ 8, 2 ]
16
17 A1 = np.array([[7, 2],
18                [3, 4]], dtype=float)
19
20 A2 = np.array([[1, 6],
21                [8, 2]], dtype=float)
22
23
24 def phi(z):
25     return max(0.0, z)
26
27
28 def bnn_ode(t, p):
29     """
30     State vector:
31         p = [x1, y1, x2, y2]
32
33     where
34         Player 1 strategy = (x1, y1)
35         Player 2 strategy = (x2, y2)
36     """
37     x1, y1, x2, y2 = p
38
39     # Expected payoff under current mixed strategies
40     u1_mix = (
41         7 * x1 * x2 + 2 * x1 * y2 +
42         3 * y1 * x2 + 4 * y1 * y2
43     )
44
45     u2_mix = (
46         1 * x1 * x2 + 6 * x1 * y2 +
47         8 * y1 * x2 + 2 * y1 * y2
48     )
49
50     # Pure-strategy payoffs against opponent's current mix
51     u11 = 7 * x2 + 2 * y2
```

```

52     u12 = 3 * x2 + 4 * y2
53
54     u2A = 1 * x1 + 8 * y1
55     u2B = 6 * x1 + 2 * y1
56
57     phi11 = phi(u11 - u1_mix)
58     phi12 = phi(u12 - u1_mix)
59     phi2A = phi(u2A - u2_mix)
60     phi2B = phi(u2B - u2_mix)
61
62     # BNN dynamics
63     dx1_dt = phi11 - x1 * (phi11 + phi12)
64     dy1_dt = phi12 - y1 * (phi11 + phi12)
65
66     dx2_dt = phi2A - x2 * (phi2A + phi2B)
67     dy2_dt = phi2B - y2 * (phi2A + phi2B)
68
69     return [dx1_dt, dy1_dt, dx2_dt, dy2_dt]
70
71
72 # Initial condition
73
74 p0 = [0.5, 0.5, 0.5, 0.5]
75
76 # Time interval
77 t_span = (0, 50)
78 t_eval = np.linspace(t_span[0], t_span[1], 2000)
79
80 # Solve ODE
81 sol = solve_ivp(
82     bnn_ode,
83     t_span,
84     p0,
85     t_eval=t_eval,
86     method="RK45",
87     rtol=1e-9,
88     atol=1e-11
89 )
90
91 # Extract solution
92 t = sol.t
93 x1, y1, x2, y2 = sol.y
94
95 # Print final state
96
97 print("Final approximate strategy profile:")
98 print(f"Player 1: x1 = {x1[-1]:.6f}, y1 = {y1[-1]:.6f}, sum = {x1[-1] + y1[-1]:.6f}")
99 print(f"Player 2: x2 = {x2[-1]:.6f}, y2 = {y2[-1]:.6f}, sum = {x2[-1] + y2[-1]:.6f}")
100
101 # Plot trajectories
102
103 plt.figure(figsize=(10, 6))
104 plt.plot(t, x1, label=r"$x^1$")
105 plt.plot(t, y1, label=r"$y^1$")

```

```
106 plt.plot(t, x2, label=r"$x^2$")
107 plt.plot(t, y2, label=r"$y^2$")
108 plt.xlabel("t")
109 plt.ylabel("Probability")
110 plt.title("BNN dynamics for the 2x2 game")
111 plt.legend()
112 plt.grid(True)
113 plt.show()
```